

CSA1522-CLOUD COMPUTING

A Comparative Study of IaaS, PaaS, and SaaS for an Educational Institution

Presented by

ARAVIND KUMAR.S (192524210)

MAGESH.S (192525002)

BRIJESH.T (192521391)

Objective & Introduction



Objective

- Understand cloud computing service models
- Compare IaaS, PaaS, and SaaS
- Learn their applications in educational institutions
- Identify the best cloud service for different academic needs



Introduction

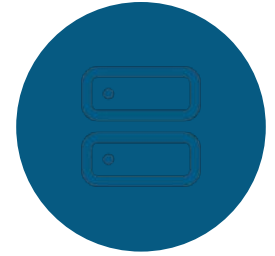
Cloud computing provides computing resources such as servers, storage, software, and applications over the Internet. It helps educational institutions reduce costs, improve accessibility, and support online learning.

3

Service Models Compared

IaaS

Infrastructure as a Service



IaaS provides virtualized computing resources such as servers, storage, and networking over the Internet.

Features

- Virtual Machines
- Cloud Storage
- Networking
- High Scalability
- Pay-as-you-use pricing

Examples

- AWS EC2
- Microsoft Azure Virtual Machines
- Google Compute Engine

Educational Use

Universities host websites, databases, and Learning Management Systems (LMS).

PaaS

Platform as a Service



PaaS provides a complete development platform for building, testing, and deploying applications.

Features

- Ready development environment
- Database support
- Automatic updates
- Easy deployment

Examples

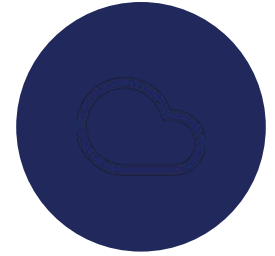
- Google App Engine
- Heroku
- Azure App Service

Educational Use

Students and faculty develop educational websites and mobile applications.

SaaS

Software as a Service



SaaS delivers ready-to-use software applications through the Internet.

Features

- No installation required
- Accessible from anywhere
- Automatic updates
- Subscription-based service

Examples

- Google Classroom
- Microsoft 365
- Gmail
- Zoom

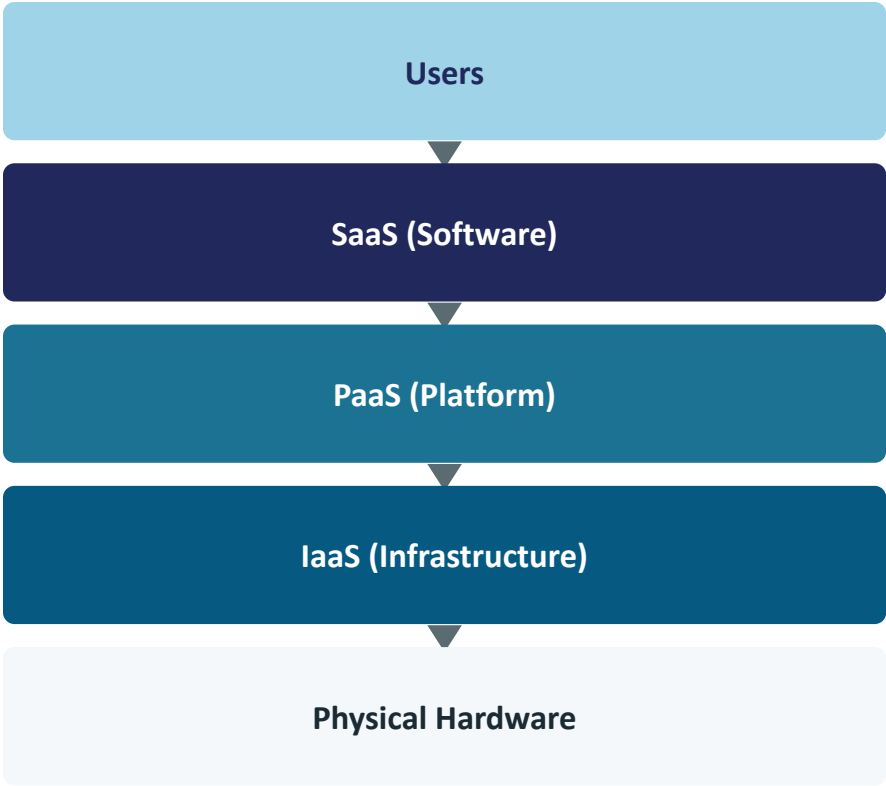
Educational Use

Online classes, assignment submission, video meetings, and collaboration.

Comparison & Architecture

Feature	IaaS	PaaS	SaaS
Provides	Infrastructure	Platform	Software
User Manages	OS & Apps	Apps	Only Usage
Technical Skill	High	Medium	Low
Cost	High	Medium	Low
Best For	Server Hosting	App Development	Online Learning

Cloud Service Stack



Real-World Use Case & Benefits

Educational Institution Use Case

IaaS

Hosts college website and LMS.

PaaS

Students develop cloud-based applications.

SaaS

Teachers conduct online classes; students submit assignments; faculty collaborate using cloud apps.

Benefits



Cost Reduction



Easy Accessibility



Data Security



Scalability



Better Collaboration



Remote Learning Support

Conclusion

Cloud computing plays a vital role in modern education. IaaS provides infrastructure, PaaS supports application development, and SaaS delivers ready-to-use software. Educational institutions can improve learning, collaboration, and resource management by choosing the right cloud service model.

Key Takeaways

IaaS → Infrastructure

PaaS → Development Platform

SaaS → Ready-to-use Software

Improves flexibility, reduces cost, and supports digital education.



Thank You



CSA1522-CLOUD COMPUTING

A Comparative Study of IaaS, PaaS, and SaaS for an Educational Institution

Presented by
ARAVIND KUMAR.S (192524210)
MAGESH.S (192525002)
BRIJESH.T (192521391)

02:38 PM | 23 Jul 2026
Thursday
Chennai, Tamil Nadu, India 🇮🇳

Poonamallee Bypass Road, Thandalam Sriperumbudur, Chennai, Tamil Nadu 602105, India
Lat 13.025715, Long 80.015436

GPS Map Camera

